

Unit I

- **Descriptive Statistics and Data Visualization**
- Introduction to statistics and its importance in engineering. Types of data: categorical, numerical, discrete, continuous. Measures of central tendency: mean, median, mode. Measures of dispersion: range, variance, standard deviation. Exploratory data analysis (EDA) techniques. Data visualization techniques: histograms, box plots, scatter plots.

THE ENGINEERING METHOD AND STATISTICAL THINKING

- Engineers solve problems of interest to society by the efficient application of scientific principles.
- The **engineering** or **scientific method** is the approach to formulating and solving these problems.

The steps in the engineering method

1. Develop a clear and concise description of the problem.
2. Identify, at least tentatively, the important factors that affect this problem or that may play a role in its solution.
3. Propose a model for the problem, using scientific or engineering knowledge of the phenomenon being studied. State any limitations or assumptions of the model.
4. Conduct appropriate experiments and collect data to test or validate the tentative model or conclusions made in steps 2 and 3.
5. Refine the model on the basis of the observed data.
6. Manipulate the model to assist in developing a solution to the problem.
7. Conduct an appropriate experiment to confirm that the proposed solution to the problem is both effective and efficient.
8. Draw conclusions or make recommendations based on the problem solution.

Statistics

- The field of **statistics** involves the collection, presentation, analysis, and use of data to make decisions and solve problems.
- **Statistics is the science of data.**
- Many aspects of engineering practice involve collecting, working with, and using data in the solution of a problem, so knowledge of statistics is just as important to the engineer as knowledge of any of the other engineering sciences. Statistical methods are a powerful aid in model verification, designing new products and systems, improving existing designs, and designing, developing, and improving production operations.

Descriptive Statistics

- Summarizing data collected about a group to describe or reach conclusions about that group only is called descriptive statistics

Inferential Statistics

- Data gathered from a sample and using the statistics generated to reach conclusion about the population from which the sample was taken, the statistics is called inferential statistics.

Population, Sample and Simple Random Sample

- In the engineering environment, the data are almost always a sample that has been selected from some population.
- A population is the entire collection of objects or outcomes about which data are collected.
- A sample is a subset of the population containing the observed objects or the outcomes and the resulting data.
- A simple random sample (SRS) of size n is a sample that has been selected from a population in such a way that each possible sample of size n has an equally likely chance of being selected.

Question

An electrical engineer measures the current flowing through a simple circuit several times and observes that the current measurements are different each time. Can we view these measurements as a simple random sample? What is the population?

Answer

- If the circuit is the same each time the measurement is made, and if the characteristics of the ammeter are unchanged, then we can view the current measurements as a simple random sample. The population is conceptual—it consists of all of the current measurements that could be made on this circuit with this ammeter.

Types of Data and Scales of Measurement

- In Statistics, the variables or numbers are defined and categorised using different **scales of measurements**.
- Each **level of measurement** scale has specific properties that determine the various use of statistical analysis.
- We will learn four types of scales such as nominal, ordinal, interval and ratio scale

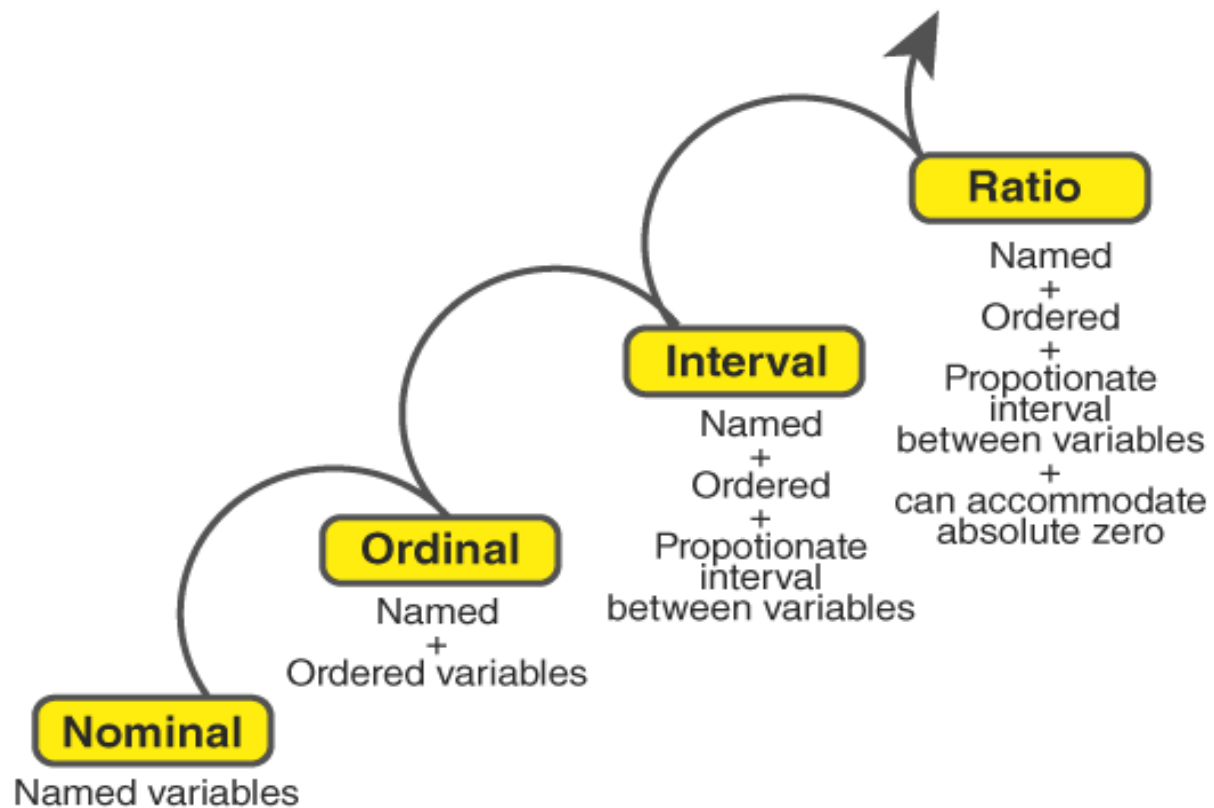
What is a Scale?

- A scale is a device or an object used to measure or quantify any event or another object.

Levels of Measurements

- There are four different scales of measurement. The data can be defined as being one of the four scales. The four types of scales are:
 - Nominal Scale
 - Ordinal Scale
 - Interval Scale
 - Ratio Scale

LEVELS OF MEASUREMENT



Nominal Scale

- A nominal scale is the 1st level of measurement scale in which the numbers serve as “tags” or “labels” to classify or identify the objects. A nominal scale usually deals with the non-numeric variables or the numbers that do not have any value.
- **Characteristics of Nominal Scale**
 - A nominal scale variable is classified into two or more categories. In this measurement mechanism, the answer should fall into either of the classes.
 - It is **qualitative**. The numbers are used here to identify the objects.
 - The numbers don't define the object characteristics. The only permissible aspect of numbers in the nominal scale is “counting.”

Nominal Scale Measurement

- **Example:**
- An example of a nominal scale measurement is given below:
- What is your gender?
- M- Male
- F- Female
- Here, the variables are used as tags, and the answer to this question should be either M or F.

Ordinal Scale

- The ordinal scale is the 2nd level of measurement that reports the ordering and ranking of data without establishing the degree of variation between them. Ordinal represents the “order.” Ordinal data is known as qualitative data or categorical data. It can be grouped, named and also ranked.
- **Characteristics of the Ordinal Scale**
 - The ordinal scale shows the relative ranking of the variables
 - It identifies and describes the magnitude of a variable
 - Along with the information provided by the nominal scale, ordinal scales give the rankings of those variables
 - The interval properties are not known
 - The surveyors can quickly analyse the degree of agreement concerning the identified order of variables

Examples of Ordinal Scale

- Ranking of school students – 1st, 2nd, 3rd, etc.
- Ratings in restaurants
- Evaluating the frequency of occurrences
 - Very often
 - Often
 - Not often
 - Not at all
- Assessing the degree of agreement
 - Totally agree
 - Agree
 - Neutral
 - Disagree
 - Totally disagree

Interval Scale

- The interval scale is the 3rd level of measurement scale.
- It is defined as a quantitative measurement scale in which the difference between the two variables is meaningful.
- In other words, the variables are measured in an exact manner, not as in a relative way in which the presence of zero is arbitrary.

Characteristics of the Interval Scale

- The interval scale is quantitative as it can quantify the difference between the values
- It allows calculating the mean and median of the variables
- To understand the difference between the variables, you can subtract the values between the variables
- The interval scale is the preferred scale in Statistics as it helps to assign any numerical values to arbitrary assessment such as feelings, calendar types, etc.

Examples the Interval Scale

- Likert Scale
- Net Promoter Score (NPS)
- Bipolar Matrix Table

Ratio Scale

- The ratio scale is the 4th level of measurement scale, which is quantitative. It is a type of variable measurement scale. It allows researchers to compare the differences or intervals. The ratio scale has a unique feature. It possesses the character of the origin or zero point

Characteristics of Ratio Scale

- Ratio scale has a feature of absolute zero
- It affords unique opportunities for statistical analysis. The variables can be orderly added, subtracted, multiplied, divided. Mean, median, and mode can be calculated using the ratio scale.
- Ratio scale has unique and useful properties. One such feature is that it allows unit conversions like kilogram – calories, gram – calories, etc.

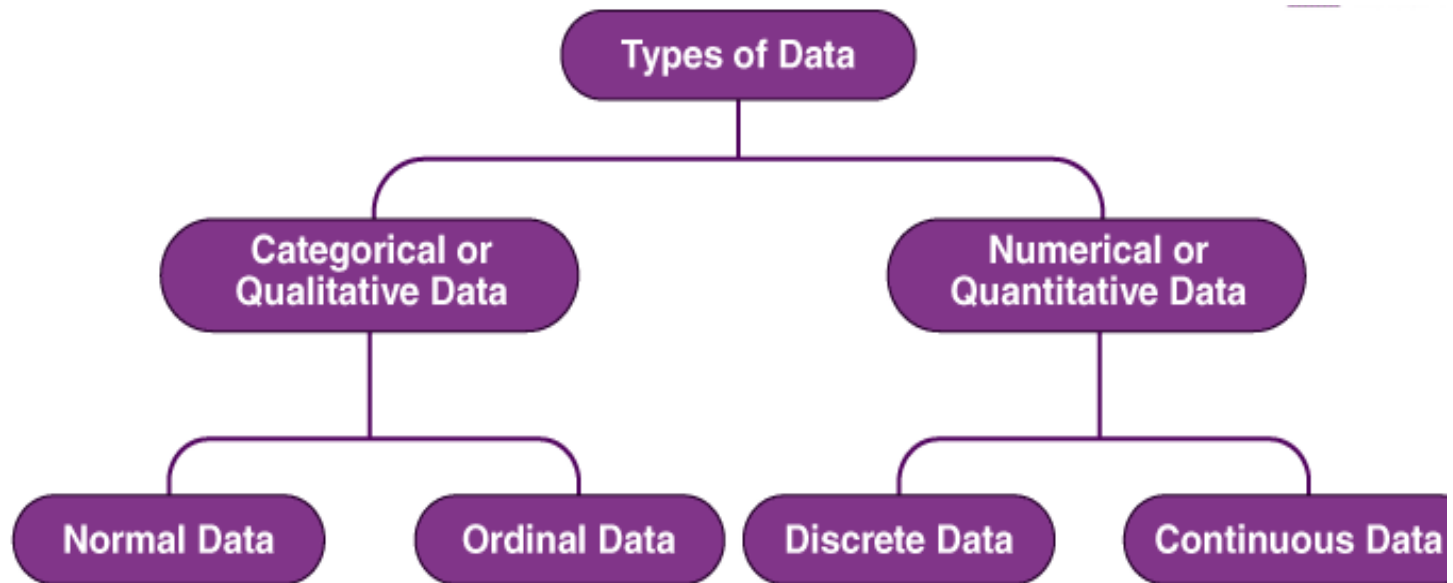
Scale Types Summary

Scale Types	Characteristics
Nominal	Entities are classified. No arithmetic meaningful.
Ordinal	Entities are classified and ordered. Cannot use + or -.
Interval	Entities classified, ordered, and differences between them understood ('units'). No zero, but can use ordinary arithmetic on intervals.
Ratio	Zeros, units, ratios between entities. All arithmetic.
Absolute	Counting; only one possible measure. All arithmetic.

Summary of Meaningful Statistics

Scale Type	Average	Spread
Nominal	Mode	Frequency
Ordinal	Median	Percentile
Interval	Arithmetic mean	Standard deviation
Ratio	Geometric mean	Coefficient of variation
Absolute	Any	Any

Types of Data



Categorical or Qualitative Data

- The categorical data consists of categorical variables which represent the characteristics such as a person's gender, hometown etc. Categorical measurements are expressed in terms of natural language descriptions, but not in terms of numbers. Sometimes categorical data can take numerical values, but those numbers do not have mathematical meaning. Some of the examples of the categorical data are as follows:
 - Birthdate
 - Favourite sport
 - School Postcode
 - Travel method to school.
- When you observe the above example, birthdate and postcode contain numbers. Even though it contains numerals, it is considered as categorical data.
- The easy way to determine whether the given data is categorical or numerical data is to try to calculate the average.
- If you cannot calculate the average, then it is considered to be a categorical data.
- Like the example mentioned above, the average of birthdate and the postal code has no meaning, so it is taken as categorical data.

Types of Categorical Data

- In general, categorical data has values and observations which can be sorted into categories or groups. The best way to represent these data is [bar graphs](#) and pie charts. Categorical data are further classified into two types namely,
 - Nominal Data
 - Ordinal Data

Categorical Variables

- In statistics, a categorical variable is a variable that contains limited, and usually a fixed number of possible values. They take values which are normally names or labels. Examples are:
 - The colour of a wall, like red, blue, pink, green, etc.,
 - Gender of people, like male, female and transgender
 - Blood group of a person: A, B, O, AB, etc.,
- These variables are used to assign each individual or another unit of observation to a particular group or nominal category based on some qualitative property.
- Generally, each of the potential values of a categorical variable is said to be as a level.
- The probability distribution linked with a random categorical variable is known as categorical distribution.

Quantitative or Numerical Data

- Quantitative data is also known as numerical data which represents the numerical value (i.e., how much, how often, how many).
- Numerical data gives information about the quantities of a specific thing. Some examples of numerical data are height, length, size, weight, and so on.
- The two different classifications of numerical data are discrete data and continuous data.

Discrete Data

- Discrete data can take only discrete values. Discrete information contains only a finite number of possible values. Those values cannot be subdivided meaningfully. Here, things can be counted in whole numbers.
- **Example:** Number of students in the class

Continuous Data

- Continuous data is data that can be calculated. It has an infinite number of probable values that can be selected within a given specific range.

Ungrouped and Grouped Data

- Raw data that has not been summarized in any way is referred to as **ungrouped data**.
- Data that has been **organized into a frequency distribution** is referred to as **grouped data**.

Data on Unemployment rates

2.3	7.0	6.3	11.3	9.6
2.8	7.1	5.6	10.6	9.1
3.6	5.9	5.4	9.7	8.3
2.4	5.5	7.1	8.8	7.6
2.9	4.7	7.1	7.8	6.8
3.0	3.9	8.0	7.5	7.2
4.6	3.6	8.4	8.1	7.7
4.4	4.1	7.5	10.3	7.6
3.4	4.8	7.5	11.2	7.2
4.6	4.7	7.6	11.4	6.8
6.9	5.9	11.0	10.4	6.3
6.0	6.4	12.0	9.5	6.0

Sorted Data

2.3	4.6	6.3	7.5	9.1
2.4	4.7	6.4	7.5	9.5
2.8	4.7	6.8	7.6	9.6
2.9	4.8	6.8	7.6	9.7
3	5.4	6.9	7.6	10.3
3.4	5.5	7	7.7	10.4
3.6	5.6	7.1	7.8	10.6
3.6	5.9	7.1	8	11
3.9	5.9	7.1	8.1	11.2
4.1	6	7.2	8.3	11.3
4.4	6	7.2	8.4	11.4
4.6	6.3	7.5	8.8	12

Frequency Distribution of Data

Class Interval	Frequency	Class Mid point	Relative frequency	Cumulative Frequency
1- less than 3				
2- less than 5				
5- less than 7				
7- less than 9				
9- less than 11				
11- less than 13				

Important Terms

- Class interval
- Range
- Class mid point
- Relative frequency
- Cumulative Frequency

Frequency Distribution of Data

Class Interval	Frequency	Class Mid point	Relative frequency	Cumulative Frequency
1- less than 3	4	2	0.0667	4
2- less than 5	12	4	0.2000	16
5- less than 7	13	6	0.2167	29
7- less than 9	19	8	0.3167	48
9- less than 11	7	10	0.1167	55
11- less than 13	5	12	0.0833	60

Histograms

- To construct a histogram for continuous data, we must divide the range of the data into intervals, which are usually called **class intervals**, **cells**, or **bins**.
- If possible, the bins should be of equal width to enhance the visual information in the histogram.
- The number of bins depends on the number of observations and the amount of scatter or dispersion in the data.
- A histogram that uses either too few or too many bins will not be informative.
- We usually find that between 5 and 20 bins is satisfactory in most cases and that the number of bins should increase with n .
- Choosing the number of bins approximately equal to the square root of the number of observations often works well in practice.
- Some basic statistics textbooks suggest using Sturges's rule, which sets the number of bins $= 1 + \log_2 n$ where n is the sample size.
- There are many variations of Sturges's rule.
- Computer software packages use many different algorithms to determine the number and width of bins, and some of them may not be based on Sturges's rule.

Histogram

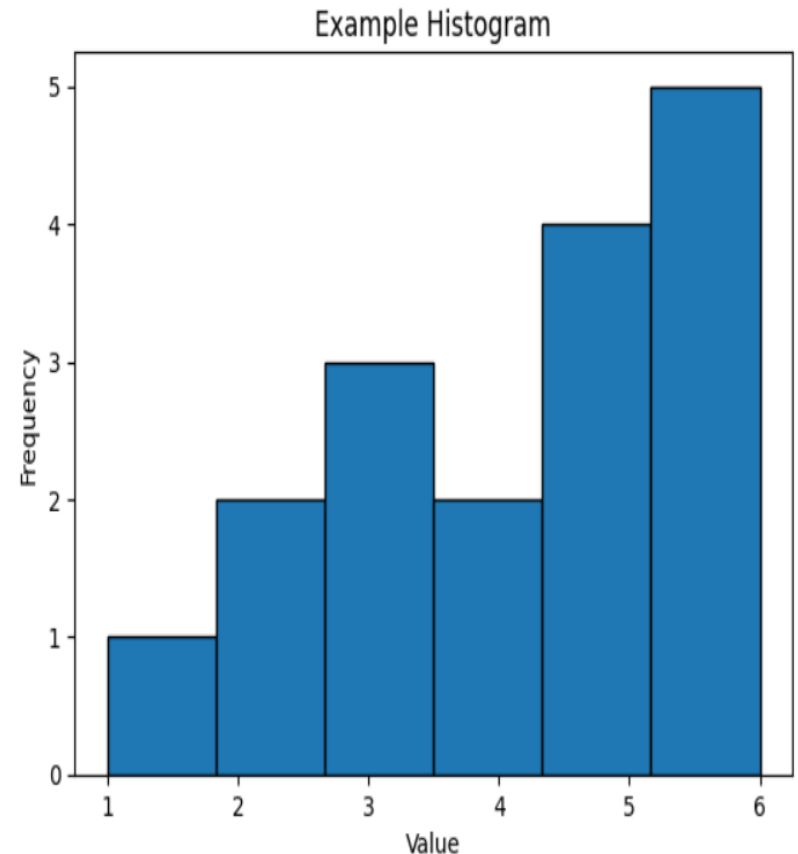
```
import matplotlib.pyplot as plt

# Sample data
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6]

# Plot
plt.hist(data, bins=6, edgecolor='black')

# Add labels and title
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Example Histogram')

# Display
plt.show()
```



Frequency Polygon

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6]

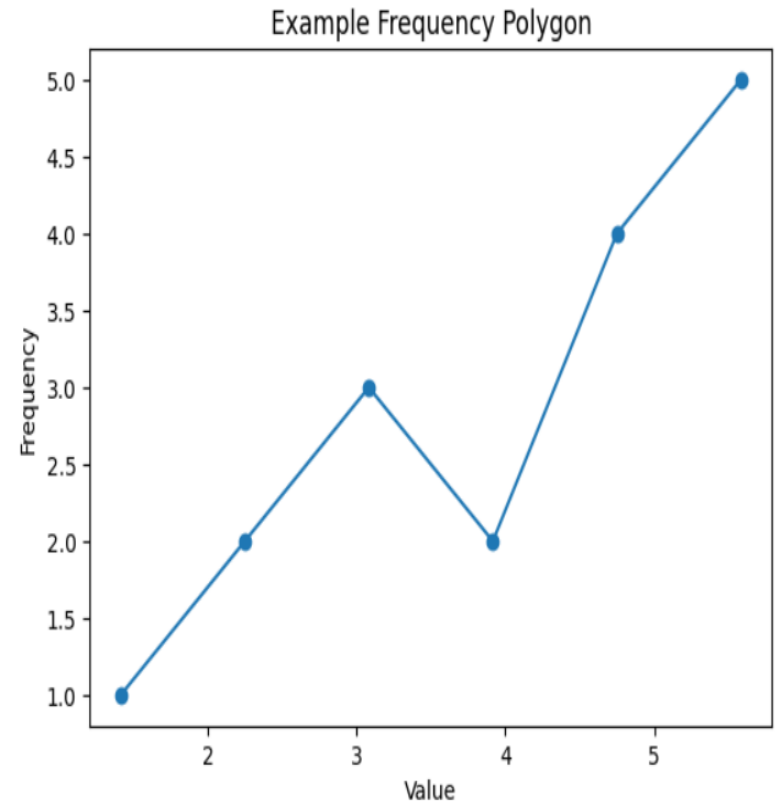
# Create a histogram to obtain the frequency data
hist, bin_edges = np.histogram(data, bins=6, range=(min(data), max(data)))

# Calculate midpoints of each bin
bin_midpoints = (bin_edges[1:] + bin_edges[:-1]) / 2

# Plot the frequency polygon
plt.plot(bin_midpoints, hist, marker='o')

# Add labels and title
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Example Frequency Polygon')

# Display
plt.show()
```



Ogive

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data
data = [1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6]

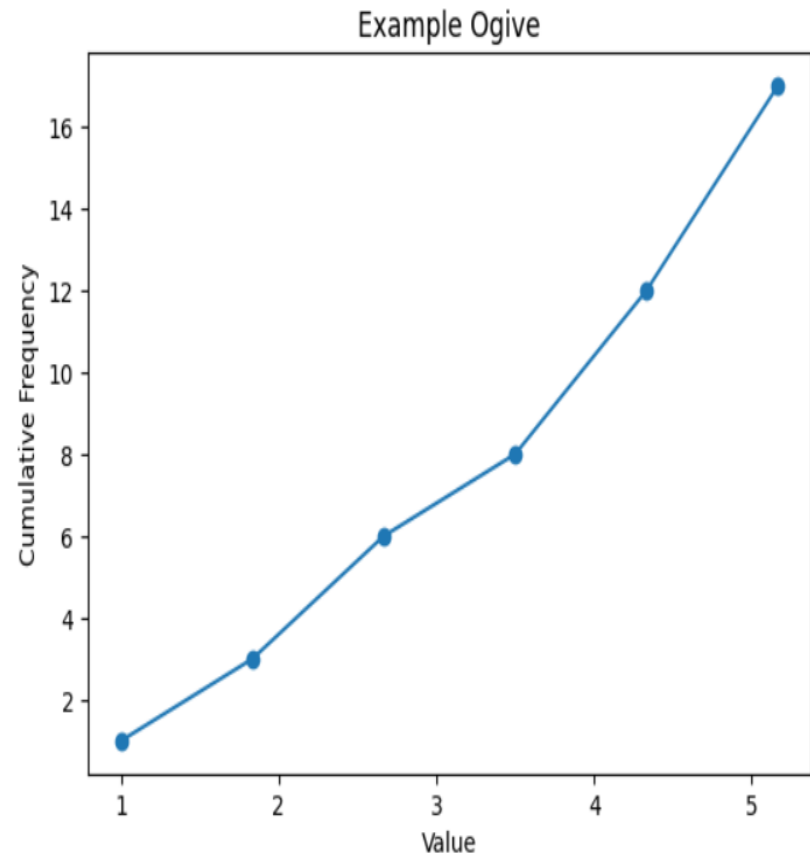
# Create a histogram to obtain the frequency data
hist, bin_edges = np.histogram(data, bins=6, range=(min(data), max(data)))

# Calculate cumulative frequencies
cumulative_freq = np.cumsum(hist)

# Plot the ogive
plt.plot(bin_edges[:-1], cumulative_freq, marker='o', linestyle='--')

# Add labels and title
plt.xlabel('Value')
plt.ylabel('Cumulative Frequency')
plt.title('Example Ogive')

# Display
plt.show()
```



Pie chart

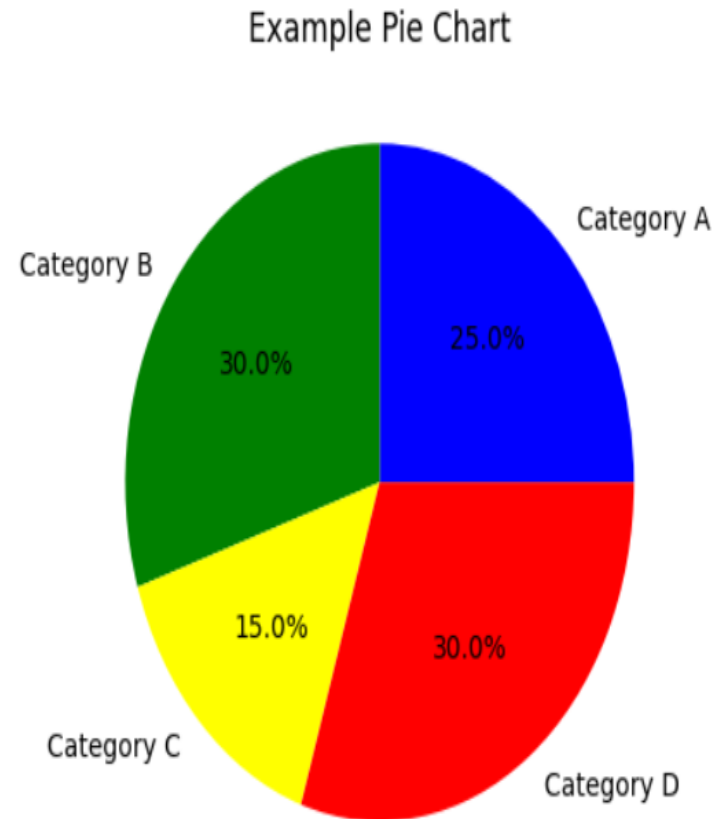
```
import matplotlib.pyplot as plt

# Sample data
labels = ['Category A', 'Category B', 'Category C', 'Category D']
sizes = [25, 30, 15, 30]
colors = ['blue', 'green', 'yellow', 'red']

# Create a pie chart
plt.pie(sizes, labels=labels, colors=colors, autopct='%1.1f%%')

# Add title
plt.title('Example Pie Chart')

# Display
plt.show()
```



Graphical Depiction

- Histograms
- Frequency Polygon
- Ogive
- Pie-chart
- Scatter Plot
- Stem and leaf plot
- Box plot

Measures of Central tendency

- Mode
- Median
- Percentile $i = P/100(n)$, P = percentile of interest, i = percentile location, n = size of dataset.
- If i is a whole number, the p th percentile is the average value at the i th location and $(i+1)$ location.
- If i is not a whole number, the p th percentile is located at the whole number part of $(i+1)$ location
- Quartiles $Q1 = P_{25}$, $Q2 = P_{50}$, $Q3 = P_{75}$
- Inter quartile range $Q3 - Q1$

Sample Mean

Sample Mean

If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$, the **sample mean** is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$= \frac{\sum_{i=1}^n x_i}{n} \quad (2-1)$$

Measures of Dispersion

Sample Variance

Sample
Variance and
Sample
Standard
Deviation

If the n observations in a sample are denoted by $x_1, x_2, \dots, x_n$, then the sample variance is

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad (2-3)$$

The sample standard deviation, s , is the positive square root of the sample variance.

- Note that the divisor for the sample variance is the sample size minus 1, $(n - 1)$, whereas for the population variance it is the population size, N .
- If we knew the true value of the population mean, μ , we could find the *sample* variance as the average squared deviation of the sample observations about μ .

- In practice, the value of μ is almost never known, and so the sum of the squared deviations about the sample average must be used instead.
- However, the observations x_i tend to be closer to their average, $\bar{x}$, than to the population mean, μ .
- Therefore, to compensate for this we use $n - 1$ as the divisor rather than n .
- If we used n as the divisor in the sample variance, we would obtain a measure of variability that is, on the average, consistently smaller than the true population variance σ .

Degrees of freedom

- Another way to think about this is to consider the sample variance s^2 as being based on $n - 1$ **degrees of freedom**.
- The term *degrees of freedom* results from the fact that the n deviations always sum to zero, and so specifying the values of any $n - 1$ of these quantities automatically determines the remaining one.

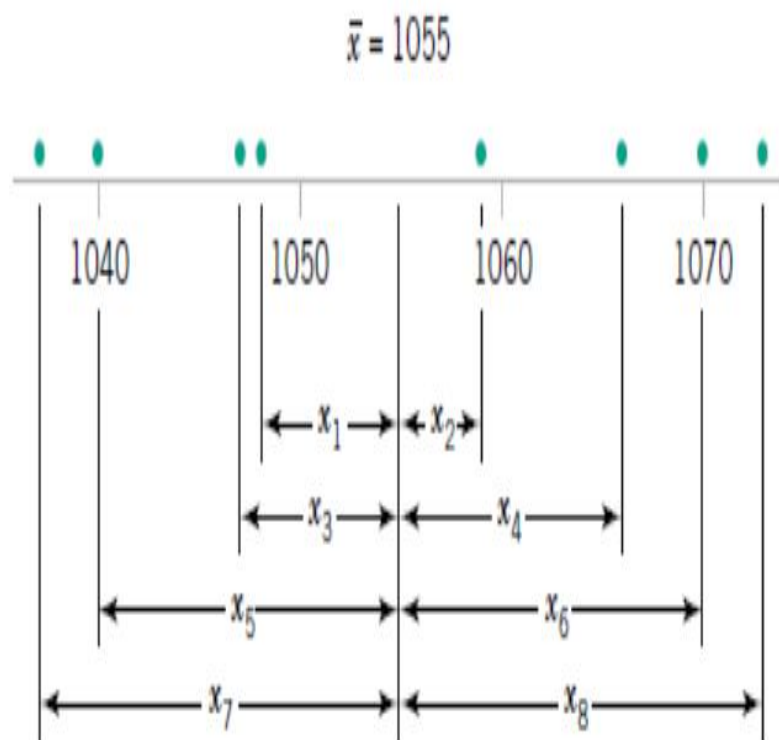


Figure 2-3 How the sample variance measures variability through the deviations $x_i - \bar{x}$.

Sample Standard Deviation

i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	1048	-7	49
2	1059	4	16
3	1047	-8	64
4	1066	11	121
5	1040	-15	225
6	1070	15	225
7	1037	-18	324
8	1073	18	324
	<u>8440</u>	<u>0.0</u>	<u>1348</u>

Correlation coefficient

- The strength of a linear relationship between two variables, y and x , can be described by the **sample correlation coefficient** r .
- Suppose that we have n pairs of observations on two variables. It would be logical to say that y and x have a positive relationship if large values of y occur together with large values of x and small values of y occur with small values of x . Similarly, a negative relationship between the variables is implied if large values of y occur together with small values of x and small values of y occur with large values of x . The corrected sum of cross-products given by

Sum of Cross product


$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) / n$$

Sample correlation coefficient

Sample Correlation Coefficient

Given n pairs of data $(y_1, x_1), (y_2, x_2), \dots, (y_n, x_n)$, the sample correlation coefficient r is defined by

$$r = \frac{S_{xy}}{\sqrt{\left(\sum_{i=1}^n (x_i - \bar{x})^2\right) \left(\sum_{i=1}^n (y_i - \bar{y})^2\right)}} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \quad (2-6)$$

with $-1 \leq r \leq +1$